SAHIL GOYAL

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Scholar GitHub LinkedIn Medium

EDUCATION

Indian Institute of Technology Roorkee *2019 - 2024*
B.Sc.+M.Sc. in Applied Mathematics
Department Rank 1 (Gold Medal) Overall GPA: **9.44**/10

PUBLICATIONS

1. **MaGNeTS: Masked Generative Nested Transformers with Decode Time Scaling.** [Paper] [Poster]
   Sahil Goyal*, Debapriya Tula*, Gagan Jain, Pradeep Shenoy, Prateek Jain, Sujoy Paul [**ICML** '25]
2. **Design-o-meter: Towards Evaluating and Refining Graphic Designs.** [Paper] [Poster] [Survey]
   Sahil Goyal*, Abhinav Mahajan*, Swasti Mishra, Prateksha Udhayanan, Tripti Shukla, K J Joseph, Balaji Vasan Srinivasan. [**WACV** '25]
3. **Emotionally Enhanced Talking Face Generation.** [Paper] [Poster] [Code]
   Sahil Goyal, Shagun Uppal, Sarthak Bhagat, Yi Yu, Yifang Yin, Rajiv Ratn Shah. [**ICCV-W '23**] [**ACM MM-W '23**]
4. **Emotional talking faces: Making videos more expressive and realistic.** [Paper] [Demo]
   Sahil Goyal, Shagun Uppal, Sarthak Bhagat, Dhroov Goel, Sakshat Mali, Yi Yu, Yifang Yin, Rajiv Ratn Shah. [**ACM MM Asia '22**] [**Best Demo paper**]

EXPERIENCE

***Pre-Doctoral Researcher* at Google DeepMind** *June 2024 - Present*
- Research in **Efficient ML** including contributions to Veo and Gemini, and publication at ICML [1].

***Research Intern* at Adobe Research** *May 2023 - July 2023*
- Unified framework [2] for automated design evaluation and refinement, enabling iterative improvements for creators.

***Research Intern* at MIDAS Lab @IIITD** *Sep 2021 - Jan 2023*
- Multimodal talking face generation network [3, 4], winning **Best Demo Paper** award at ACM MM Asia.

***Machine Learning Intern* at PANDO** *March 2021 - April 2021*
- Developed a **20x** faster 3-D space optimization algorithm for transport logistics.

KEY RESEARCH PROJECTS

Compression of Pre-trained LLMs *May 2025 - Present*
Advisors: Dr. Sujoy Paul, Dr. Prateek Jain, Dr. Pradeep Shenoy
- Optimizing **model compression** pipeline for Gemini, aiming for **minimal extra training cost**.
- Building and refining effective **structured-pruning** and **layer-wise distillation** techniques.
- Initial results show significant quality gains over iso-params and iso-FLOPs baselines.

Adaptive Compute for Visual Generation *June 2024 - Present*
Advisors: Dr. Sujoy Paul, Dr. Prateek Jain, Dr. Aditya Kusupati
- Developed an **adaptive computation** method [1] for multi-step generative models using **nested transformers**.
- Achieved ∼**3x** reduction in inference FLOPs for image/video generation.
- Implemented similar efficient techniques for Google's SOTA video generation model (Veo).
- Extending the method for **on-device** generation using **recursive fixed-point transformers**.

Automated Evaluation and Refinement of Graphic Designs *May 2023 - July 2023*
Advisors: Dr. Joseph KJ, Dr. Balaji Vasan Srinivasan
- Built compact **Siamese** networks (**0.5M** params) to score Graphic Designs.

- Significantly outperformed GPT-4o and LLaVA-NeXT (7B) in ranking graphic designs.
- Tailored advanced **Genetic Algorithms** for SOTA **Layout refinement** (mean-IOU of **0.54** on Crello dataset).

Emotionally Enhanced Talking Face Generation *Sep 2021 - Jan 2023*

Advisors: Dr. Rajiv Ratn Shah

- Developed the first **talking face generation** network [3, 4] that incorporated emotions.
- SOTA results on **CREMA-D** dataset (**83.2%** emotion classification accuracy and **FID** metric: **5.29**).
- Deployed the entire system as an interactive web interface.

SELECTED ACHIEVEMENTS

- Among **15**, from 18,000+ applicants, selected for Google's Pre-Doctoral Researcher Program. *2024*
- **Department Gold Medal** (IIT Roorkee - Applied Mathematics). *2024*
- **Best Demo Paper** at ACM MM ASIA conference. *2022*
- **Gold Medal** at Inter IIT Tech Meet. *2022*
- **Open Source**: PyTorch-Vision (debugged dataloaders, ported tests and updated docs). *2022*
- Awarded ACM SigMM student travel grant for ACM MM Asia. *2022*
- JEE Advanced Rank of **1583** among 1.5M students. *2019*
- **600+** stars on GitHub.

PROJECTS B.C. (BEFORE CHATGPT)

Model Extraction Attacks for Video Classification [Code] *Feb 2022 - March 2022*

- Successfully extracted models across grey-box (5% original data) and black-box (synthetic data only) settings.
- Developed novel data augmentation, synthetic data generation and knowledge distillation techniques.
- Our method won **gold medal** at Inter IIT Tech Meet.

Global Wheat Detection Challenge 2021 [Code] *June 2021 - July 2021*

- Implemented and benchmarked SOTA **object detection** models with diverse backbones for wheat-head detection.
- Improved accuracy to **56.2%** using training tweaks (data augmentation, pseudo labeling) and post-processing techniques (weighted boxes fusion, non-maximum suppression).

Multimodal Image-to-Image Translation [Code] [Blog] *Dec 2020 - Jan 2021*

- Implemented **BicycleGAN** from scratch in PyTorch for **edges-to-images** and **denoising images** tasks.
- Reproduced the results for paper's diversity-based **LPIPS** benchmark.

VisualML [Code] [Demo] *May 2020 - July 2020*

- Developed a set of interactive, **browser-based ML visualization** demos.
- Engineered in-browser execution by integrating TensorFlow.js, Plotly.js, and numeric libraries.

POSITIONS OF RESPONSIBILITY

- **Reviewer** at CVPR, ICML, ICLR, ICCV, AAMAS, WACV conferences. *2024 - Present*
- **Mentor** at Data Science Group IITR *2020 - 2024*
- **Undergraduate Teaching Assistant** at Academic Reinforcement Program IITR *2021*
- **Teaching Volunteer** at National Social Service IITR *2019 - 2020*

RELEVANT KEY COURSES

Machine Learning Stanford CS229 (Machine Learning), Stanford CS231n (Computer Vision), Deep Learning Specialization, Data Mining, CVIT Summer School of AI.

CS and Maths Data Structures and Algorithms, Linear Algebra, Mathematical Statistics, Graph Theory, Statistical Inference, Parallel Computing, Discrete Mathematics, Numerical Methods, Real Analysis, Functional Analysis, Topology, Combinatorics, Differential Equations